Interop QOS

2026-03-23-12-43-40

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions. By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels, ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

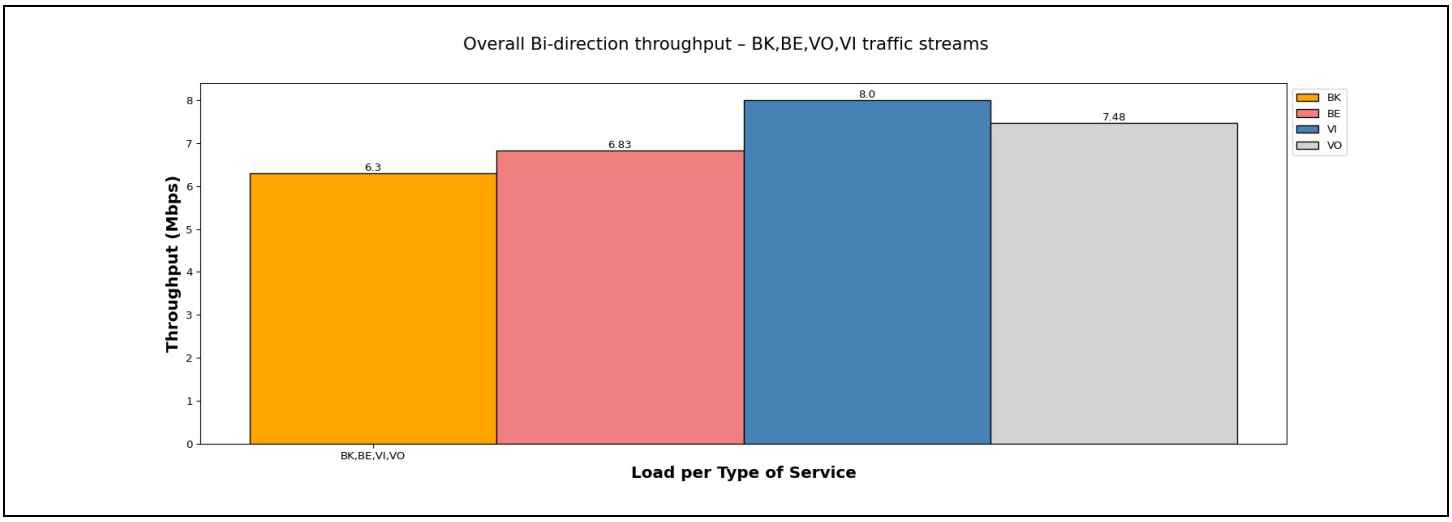
Test Configuration	Device List	vivonew(Android), ubuntu24(Linux), macoss-MacBook-Air.local(Mac), sungsung(Android)
	Number of Stations	Total(4) Android(2) Linux(1) Mac(1)
	AP Model	Cisco
	SSID	NETGEAR_5G_wpa2
	Traffic Duration in hours	0.02
	Security	wpa2
	Protocol	UDP
	Traffic Direction	Bi-direction
	TOS	['BK', 'BE', 'VI', 'VO']
	Per TOS Load in Mbps	Upload:1.0 Mbps,Download:1.0 Mbps

Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 1.0 Mbps-upload	Throughput for Load 1.0 Mbps-download
4	BK : 3.15, BE : 3.41, VI: 4.0, VO: 3.74	BK : 3.15, BE : 3.42, VI: 4.0, VO: 3.74

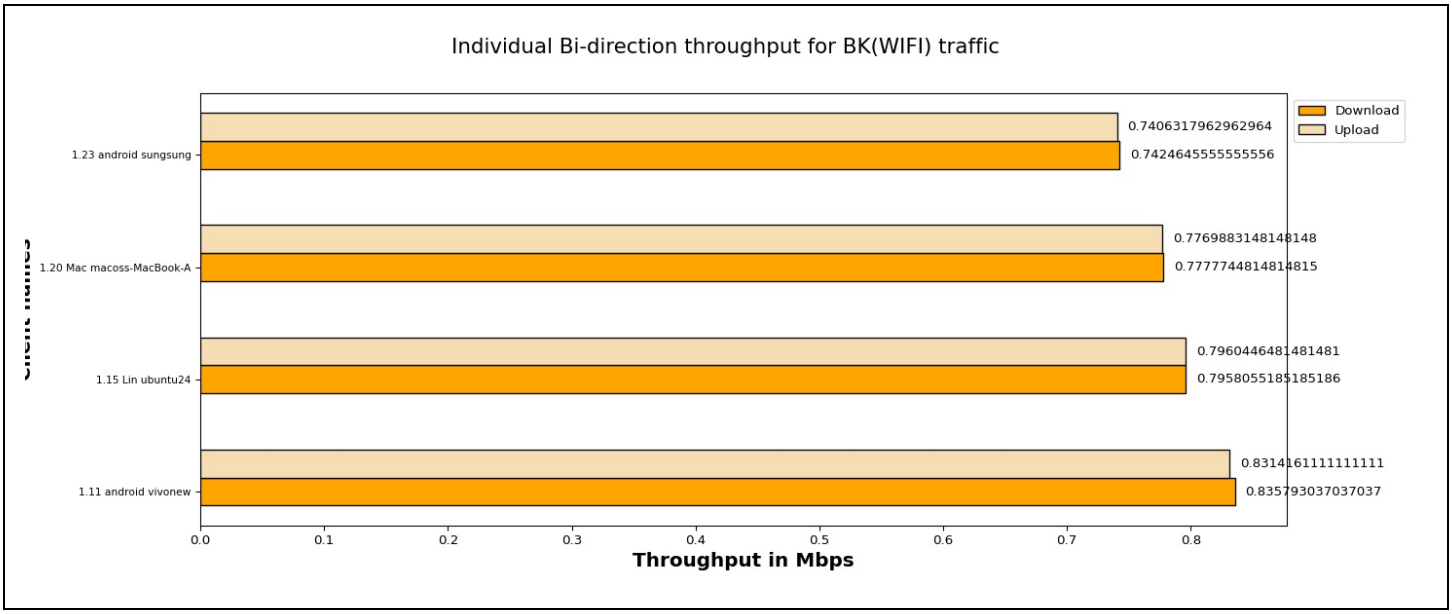
Overall Bi-direction throughput for 4 clients with different TOS.

The below graph represents overall Bi-direction throughput for all connected stations running BK, BE, VO, VI traffic with different intended loadsUpload:1.0 Mbps,Download:1.0 Mbps per tos



Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BK(WiFi).

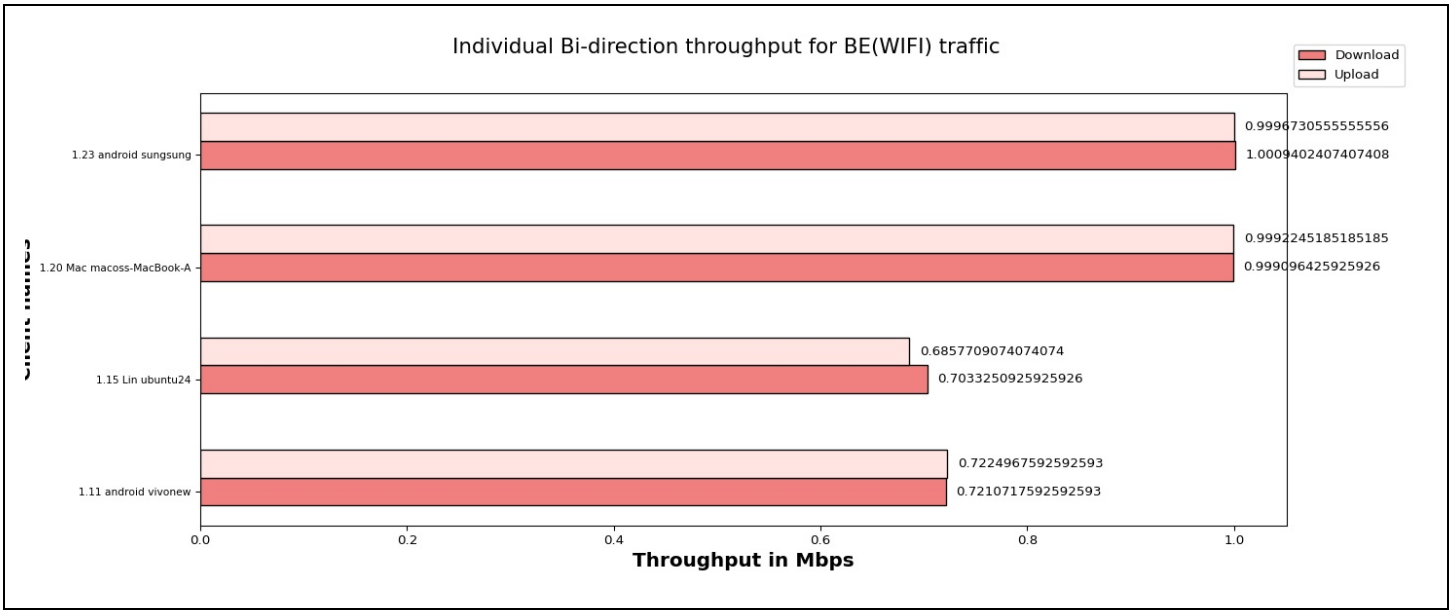
The below graph represents individual throughput for 4 clients running BK (WiFi) traffic. X- axis shows “Throughput in Mbps” and Y-axis shows “number of clients”.



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	NETGEAR_5G_wpa2	Background	1.0 Mbps	1.0 Mbps	0.8314161111111111 Mbps	0.835793037037037 Mbps	0.008139	0.009294
1.15 Lin ubuntu24	c:5f:67:8c:93:86	NETGEAR_5G_wpa2	Background	1.0 Mbps	1.0 Mbps	0.7960446481481481 Mbps	0.7958055185185186 Mbps	0.001846	0.030775
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Background	1.0 Mbps	1.0 Mbps	0.7769883148148148 Mbps	0.7777744814814815 Mbps	0.000634	0.025337
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Background	1.0 Mbps	1.0 Mbps	0.7406317962962964 Mbps	0.7424645555555556 Mbps	0.000000	0.034224

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 4 clients running BE (WiFi) traffic. X- axis shows “number of clients” and Y-axis shows “Throughput in Mbps”.

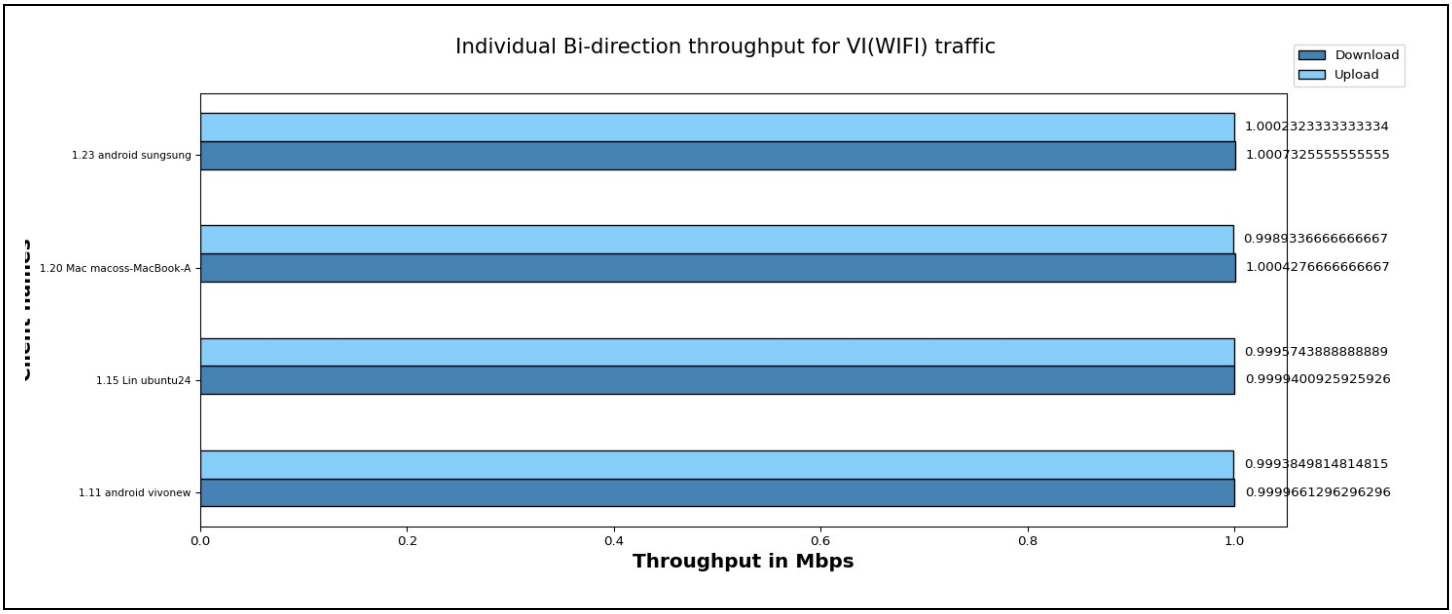


Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	NETGEAR_5G_wpa2	Besteffort	1.0 Mbps	1.0 Mbps	0.7224967592592593 Mbps	0.7210717592592593 Mbps	0.000000	0.008495

1.15 Lin ubuntu24	c:5f:67:8c:93:86	NETGEAR_5G_wpa2	Besteffort	1.0 Mbps	1.0 Mbps	0.6857709074074074 Mbps	0.7033250925925926 Mbps	0.000742	0.000669
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Besteffort	1.0 Mbps	1.0 Mbps	0.9992245185185185 Mbps	0.999096425925926 Mbps	0.001017	0.010242
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Besteffort	1.0 Mbps	1.0 Mbps	0.9996730555555556 Mbps	1.0009402407407408 Mbps	0.000506	0.006838

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VI(WiFi).

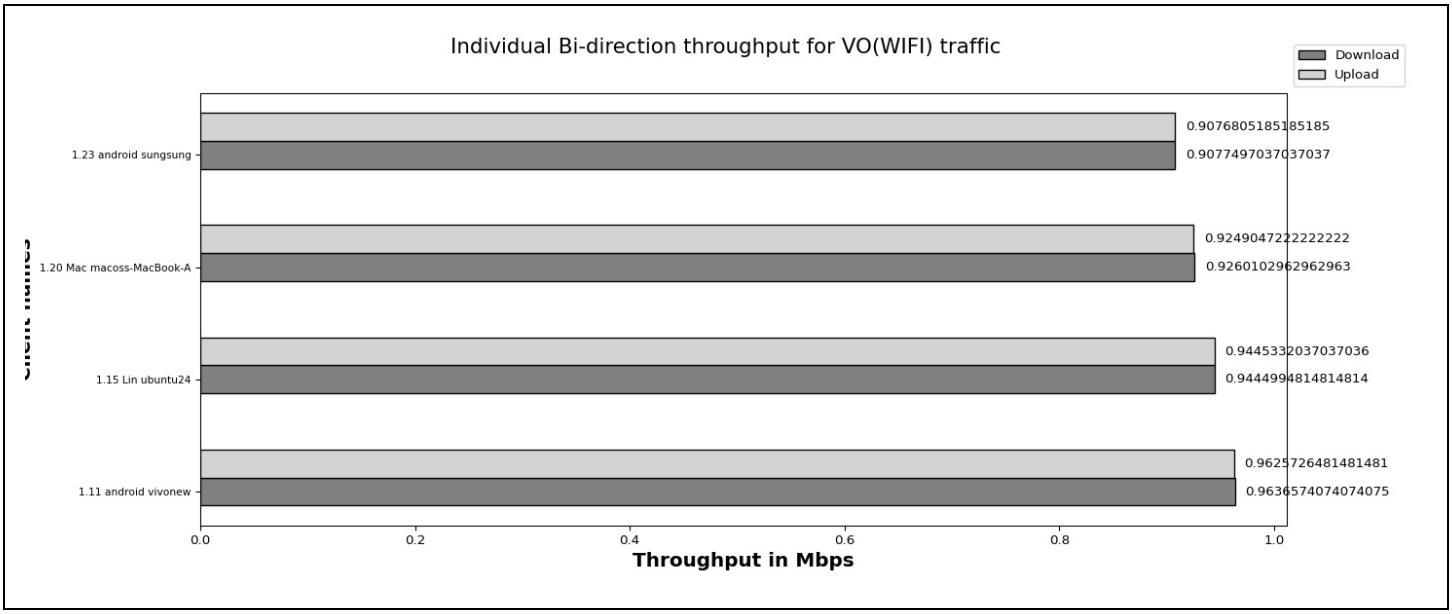
The below graph represents individual throughput for 4 clients running VI (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	NETGEAR_5G_wpa2	Video	1.0 Mbps	1.0 Mbps	0.9993849814814815 Mbps	0.9999661296296296 Mbps	0.009283	0.010254
1.15 Lin ubuntu24	c:5f:67:8c:93:86	NETGEAR_5G_wpa2	Video	1.0 Mbps	1.0 Mbps	0.9995743888888889 Mbps	0.9999400925925926 Mbps	0.000000	0.002785
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Video	1.0 Mbps	1.0 Mbps	0.9989336666666667 Mbps	1.0004276666666667 Mbps	0.006115	0.005847
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Video	1.0 Mbps	1.0 Mbps	1.0002323333333334 Mbps	1.0007325555555555 Mbps	0.003897	0.016095

Individual Bi-direction throughput with intended load Upload:1.0 Mbps,Download:1.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 4 clients running VO (WiFi) traffic. X- axis shows "number of clients" and Y-axis shows "Throughput in Mbps".



Client Name	MAC	SSID	Type of traffic	Offered upload rate	Offered download rate	Observed average upload rate	Observed average download rate	Observed Upload Drop (%)	Observed Download Drop (%)
1.11 android vivonew	92:da:84:1a:a3:75	NETGEAR_5G_wpa2	Voice	1.0 Mbps	1.0 Mbps	0.9625726481481481 Mbps	0.9636574074074075 Mbps	0.016540	0.004041
1.15 Lin ubuntu24	c:5f:67:8c:93:86	NETGEAR_5G_wpa2	Voice	1.0 Mbps	1.0 Mbps	0.9445332037037036 Mbps	0.9444994814814814 Mbps	0.000883	0.003315
1.20 Mac macoss-MacBook-Air.local	30:35:ad:c5:e1:be	NETGEAR_5G_wpa2	Voice	1.0 Mbps	1.0 Mbps	0.9249047222222222 Mbps	0.9260102962962963 Mbps	0.000000	0.002470
1.23 android sungsung	ca:84:bc:fa:cb:7a	NETGEAR_5G_wpa2	Voice	1.0 Mbps	1.0 Mbps	0.9076805185185185 Mbps	0.9077497037037037 Mbps	0.000000	1.865917

Information	contact	support@candelatech.com